

Media Bus: XR-Based Immersive Cultural Heritage Tourism

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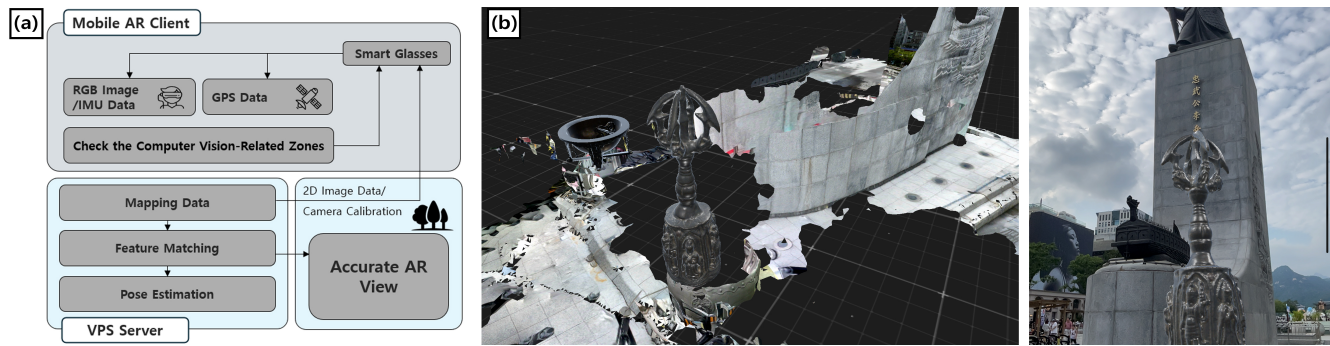


Figure 1: (a) System Architecture Overview (b) Consistent AR Content Augmentation using VPS

Abstract

This study presents Media Bus prototype for facilitating digital storytelling experiences through extended reality (XR), head mounted displays (HMDs), and transparent OLEDs within the city of Seoul. The system offers an immersive tourism experience by integrating precise location tracking, which combines Visual Positioning Service (VPS) and Global Positioning System (GPS), along with TOLED displays and wearable augmented reality (AR) content. Two rounds of expert interviews were conducted to guide the development and assess the prototype. The preliminary interview provided foundational concepts and technical requirements, while the subsequent interview focused on evaluating the prototype and suggesting potential improvements. The results indicate that the proposed system has significant potential for enhancing urban tourism experiences in an innovative manner. However, challenges remain in improving location accuracy and refining user interfaces. This research presents a novel application of XR and is anticipated to advance smart city initiatives and cultural tourism efforts.

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1 Introduction

Regions with heritage sites are developing various strategies to offer rich, immersive experiences for both tourists and local residents. Technological advancements, particularly in digital storytelling, play a pivotal role in engaging visitors [Mudička and Kapica 2022].

Recent studies investigate methods to enhance interactive experiences at historical sites using XR technology. The importance of attention-guiding mechanisms in AR for integrating virtual content with physical spaces and supporting multi-episode narratives was emphasized [Zhao et al. 2021]. A novel approach, Affectual Dramaturgy, combines AR with immersive theater techniques to create immersive AR-based performances at cultural heritage sites [Dima and Maples 2021]. These research trends are advancing towards providing immersive experiences of cultural heritage in outdoor environments, extending beyond traditional indoor settings.

To stimulate public interest and engagement by improving outdoor tracking accuracy, VPS system has been used for the cultural heritage tour of the Euston Arch in London, UK [Lu et al. 2023], while AR systems combined with GPS and physical markers have been developed for cultural heritage tourism in Columbia Sucre [Salgado et al. 2021]. A location-based mobile AR application using GNSS and IMU was developed for La Laguna in the Canary Islands, but user testing revealed synchronization issues between virtual and real views [Sánchez Berriel et al. 2023]. These studies encountered limitations in delivering dynamic experiences, often constrained by the accuracy of location-based systems.

2 Our Approach

To address these limitations and provide a more immersive heritage tourism experience, this study conducted preliminary research through expert interviews and established the following objectives:

- High-Precision Positioning System with VPS and GPS: A system that achieves sub-meter level location accuracy to ensure GPS stability and continuous tracking in urban areas.
- (1) Overlap Display with TOLED: Integrate TOLED displays with real urban landscapes.
- (2) Mobile Digital Storytelling with HMD: Develop a system that delivers synchronized content through HMD in mobile settings, aligning narratives with the surroundings.

This multi-layered approach aims to provide a highly personalized and context-aware urban exploration experience. The methodology adopts an iterative design process of two rounds of expert interviews: the first to establish initial concepts and derive technical requirements, and the second to evaluate the prototype and identify optimization directions. This research introduces a new paradigm at the intersection of location-based XR and tourism with potential broad applications in smart city initiatives and digital heritage.

3 Implementation

This system aims to enhance the user experience in cultural heritage tourism by combining GPS and VPS for more accurate AR content placement. While GPS is suitable for approximate location detection, its margin of error of about 20m limits its accuracy for precise AR placement. VPS, on the other hand, provides high accuracy but requires pre-scanning of areas for content augmentation and puts a strain on device performance due to comparison operations. To address these issues, our system first uses GPS to roughly identify the area where AR content will be augmented, then activates VPS in that specific area, reducing scanning costs and device performance degradation. We also introduced an algorithm that compares real-time camera images with VPS scan data, optimizing the system for VPS communication to occur only when the user is looking at a scanned area. This results in an efficient AR augmentation system that combines the low cost of GPS with the high accuracy of VPS.

The system operates by connecting the Nreal Light to an Android device, using GPS for the user's approximate location. Then, real-time camera images with pre-scanned VPS data pinpoint the exact location. Once both conditions are met, the system receives AR content and precise location information from the VPS server and projects it onto either the HMD or TOLED. This implementation effectively combines the low cost of GPS with the high accuracy of VPS, resulting in an efficient AR augmentation system.

3.1 AR Content with HMD

In this prototype, the Nreal Light is paired with an Android device, as shown in Figure 1. The Android device connected to the HMD uses GPS to estimate the user's approximate location, limiting unnecessary VPS server communication, and finds suitable mapping data. The device's camera then utilizes computer vision to verify if the user's view matches a pre-mapped zone. If both conditions are met, the system retrieves AR content and location information from the VPS server, which is subsequently projected into the HMD.

3.2 TOLED

A 30-inch TOLED is selected to fit the vehicle's side window and mounted on the side rear window. When AR content is not displayed, the TOLED shows content to maintain user immersion.

4 Evaluation and Discussion

The second expert evaluation of this prototype involved semi-structured interviews with six specialists in tourism, XR, Human-Computer Interaction (HCI), and digital heritage. Each interviews involved 10 questions and conducted for an hour. The experts agreed the integration of cultural heritage content with XR is innovative and likely to significantly enhance the tourism experience. Yet, they suggested improvements in real-time location-based content alignment, motion sickness, and user interface intuitiveness.

Technical evaluations indicated that the VPS-GPS fusion system achieved an average location accuracy of approximately 0.25 meters ($SD=0.04m$), though accuracy was less reliable in areas with dense high-rise buildings. While the TOLED display provided clear image quality, its visibility under strong sunlight needs improvements. Key challenges include maintaining stable location tracking in varied urban environments and optimizing real-time content rendering.

Based on these evaluation results, the Media Bus has the potential to serve as an innovative tool for replacing and/or complementing guided tours within the cultural heritage tourism sector, enabling users to gain experiential knowledge of history and culture.

Future works will focus on incorporating an Inertial Navigation System (INS) to improve location tracking accuracy [Zhu et al. 2024]. These advancements are expected to elevate the immersive experience of the Media Bus at commercialization stage.

The limitations of this study include the brief evaluation period. Future research would require field tests with user groups to more accurately assess the system's practicality and effectiveness. Furthermore, evaluations of system performance under varying weather conditions and at different times of day would be necessary.

Table 1: Demographics of Experts

ID	Age	Job Position	ID	Age	Job Position
P1	40's	XR Researcher	P4	30's	HCI Researcher
P2	40's	Professor	P5	40's	XR Researcher
P3	20's	XR Developer	P6	30's	Curator

Note: All participants have 5+ years of experience.

5 Conclusion

This study proposes a Media Bus prototype that leverages XR to enable digital storytelling experiences through HMD and TOLED in the city of Seoul. With advancements in technology and optimizations in user experience, this system is anticipated to establish a new paradigm for urban tourism.

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